

Automated Skin Cancer Detection Using Deep Learning (CNN)

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Abstract - Skin cancer is one of the most common and potentially life-threatening diseases, but early detection can significantly improve treatment outcomes. In this major project, a deep learning-based skin lesion classification system was developed to assist in the automated identification of dermoscopic images. The model was trained and evaluated on the HAM10000 skin lesion dataset, which contains seven classes of skin diseases: akiec, bcc, bkl, df, mel, nv, and vasc. To address class imbalance and improve learning on difficult minority classes, a transfer learning approach with focal loss was used. The final system processes input images, performs preprocessing and normalization, and predicts the most likely lesion category along with confidence scores. The evaluation on the held-out test set achieved an overall accuracy of approximately 74.8% with a weighted F1-score of 0.72, showing strong performance on common benign lesions and reasonable performance on malignant classes. The results demonstrate that the proposed model can serve as a useful decision-support tool for preliminary skin lesion screening and can help improve early diagnostic assistance in clinical workflows.

keywords : Skin-lesion classification, skin cancer detection, deep learning, dermoscopic image analysis, HAM10000 dataset, transfer learning, focal loss, convolutional neural network, medical image classification, automated diagnosis.

1. INTRODUCTION

Skin cancer is one of the most common and dangerous diseases affecting people around the world. The World Health Organization (WHO) reports that each year, over 2 to 3 million cases of non-melanoma skin cancer and about 132,000 cases of melanoma are diagnosed globally. In countries such as Australia and the United States, skin cancer rates have risen faster than almost any other type of cancer over the past few decades. Melanoma is especially concerning because it is the most aggressive form of skin cancer and can spread quickly to other organs if not caught early. Early detection makes a huge difference—the five-year survival rate is around 99% when melanoma is identified in its early stages, but this number can fall drastically to just 15–20% when diagnosed late.

Even though skin cancer is widespread and dangerous, diagnosing it early is still a major challenge. Traditional methods rely on visual inspection by dermatologists and

confirmation through a biopsy.

While effective, this process is slow, expensive, and heavily dependent on the doctor's experience. Many rural and underdeveloped regions also lack dermatology specialists and proper diagnostic facilities, which often leads to delayed detection and poorer outcomes. This creates a strong need for automated, reliable, and affordable detection tools that can support healthcare systems worldwide.

This major project focuses on building a deep learning-based skin lesion classification system using the HAM10000 dataset. The model is trained to identify seven different types of skin lesions from dermoscopic images: akiec, bcc, bkl, df, mel, nv, and vasc. The main goal is to help in the early prediction of lesion type by providing fast and reliable image-based classification.

The project follows a complete machine learning pipeline, including data loading, preprocessing, model training, evaluation, and prediction. The final model gives a predicted class label along with confidence scores, making it suitable for practical decision-support use. Overall, this project demonstrates how AI can assist in medical image analysis and improve the speed and consistency of preliminary skin lesion screening.

2. LITERATURE REVIEW

The application of machine learning and deep learning in medical image analysis has significantly grown over the past decade, especially in dermatological diagnostics. Numerous studies have demonstrated the potential of computer vision, particularly Convolutional Neural Networks (CNNs), to detect skin cancer with accuracy comparable to or surpassing that of human dermatologists. This literature review explores significant milestones, state-of-the-art methodologies, and technological trends related to automated skin cancer detection.

2.1 Early Developments in Skin Cancer Detection Using AI

The use of image processing and classical machine learning for skin lesion classification began in the early 2000s. Techniques such as Support Vector Machines (SVMs), Random Forests, and k-Nearest Neighbors (kNN) were employed with handcrafted features like color histograms,

texture descriptors (e.g., Gabor filters), and shape-based metrics. However, these methods were heavily reliant on manual feature extraction, limiting scalability and performance.

2.2 Landmark Studies and Contributions

2.2.1 Esteva et al. (2017)

One of the most cited studies in this field, Esteva et al. trained a deep CNN on over 129,000 clinical images, classifying skin lesions into keratinocyte carcinoma, melanoma, and benign nevi. The model achieved performance on par with 21 board-certified dermatologists, marking a paradigm shift in AI-based dermatological diagnostics.

2.2.2 Tschandl et al. (2018): HAM10000 Dataset

This study introduced the HAM10000 ("Human Against Machine with 10,000 images") dataset, which quickly became a benchmark for skin lesion classification tasks. It includes 10,015 dermoscopic images annotated across seven diagnostic categories, enabling multi-class classification and supporting AI research through open access.

2.2.3 Codella et al. (2019): ISIC Challenges

The International Skin Imaging Collaboration (ISIC) has conducted annual competitions from 2016–2020 to encourage algorithmic development in melanoma classification. These challenges provide large-scale image datasets, including metadata and lesion segmentations. Participating models often use CNNs or ensemble techniques.

2.3 CNN and Transfer Learning in Skin Lesion Classification

2.3.1 Kanchana et al. (2024): MobileNet V2-Based Classification

This study utilized MobileNet V2 variants (B0 to B7) for classifying skin lesions using transfer learning. The model achieved over 95% accuracy and reduced training time, thanks to MobileNet V2's compound scaling of depth, width, and resolution.

2.3.2 Azeem et al. (2023): SkinLesNet Architecture

A novel architecture, SkinLesNet, was designed specifically for classifying dermoscopic images. It improved generalization using dense connections and dropout regularization. SkinLesNet outperformed ResNet50 and VGG16 on the HAM10000 dataset in terms of both precision and recall.

2.3.3 Venugopal et al. (2023): Modified MobileNet-V2

This work presented a Modified MobileNet-V2-M architecture optimized for recall and sensitivity, especially important in medical applications where false negatives can be life-threatening. It incorporated attention mechanisms and adaptive learning rates, boosting melanoma classification performance.

3. PROPOSED METHODOLOGY

The proposed methodology is a deep learning-based skin lesion classification pipeline designed to take dermoscopic images as input, process them through a standardized workflow, and generate a final predicted lesion class with confidence scores. It is built around the HAM10000 dataset, a class-imbalanced medical image dataset containing seven lesion categories: akiec, bcc, bkl, df, mel, nv, and vasc. The methodology focuses on improving classification performance through preprocessing, balanced learning, transfer learning, and careful evaluation.

Each component is explained in detail below:

3.1 Data Collection and Dataset Preparation:

The first phase begins with the collection of dermoscopic images and metadata from the HAM10000 dataset. Each image is associated with diagnostic metadata such as lesion type, lesion ID, age, sex, and localization. The dataset is organized into separate training, validation, and testing splits so that the model is trained only on training data and evaluated on completely unseen test images.

The image files are stored in two main folders, and the metadata links each image id to its corresponding diagnosis label. This structure is important because it ensures that labels are consistent and traceable throughout the pipeline. Since the dataset contains multiple lesion types with uneven distribution, the class frequencies are analyzed early in the process to understand imbalance before model training starts.

3.2 Data Cleaning and Label Encoding:

Before training, the metadata is cleaned and filtered to ensure only valid records with available image paths are used. Invalid or missing image paths are removed so the final dataset contains only usable samples. The diagnosis labels are then converted from text categories into numeric form using label encoding so that the neural network can process them efficiently.

Each class is mapped to an integer, and the same mapping is used across train, validation, and test splits. This step is critical because it guarantees label consistency and prevents class mismatch during prediction. The encoded labels are later used for training, evaluation, and final classification reporting.

3.3 Image Preprocessing:

All input images are resized to a fixed spatial size of 224×224 pixels, which makes them compatible with the selected deep learning architecture. Resizing is necessary because convolutional neural networks require consistent input dimensions. After resizing, images are converted into RGB format and normalized to the range $[0, 1]$ by dividing pixel values by 255.

This normalization stabilizes training and helps the model converge faster. The images are then organized into batched datasets so that training and evaluation can be performed

efficiently. In your project, the use of a TensorFlow data pipeline allows preprocessing to happen in a structured and reproducible way.

3.4 Handling Class Imbalance:

One of the biggest challenges in the project is that the dataset is highly imbalanced. Classes such as *nv* appear much more frequently than rare classes like *df* or *vasc*. If a model is trained directly on such data, it can become biased toward the majority class and perform poorly on minority classes.

To address this, the project uses focal loss instead of standard categorical cross-entropy. Focal loss reduces the contribution of easy examples and focuses the model more on hard-to-classify samples. This is especially useful in medical diagnosis tasks where rare but clinically important classes must not be ignored. The methodology may also include class-aware sampling or class weighting depending on the training strategy used.

3.5 Model Design and Transfer Learning:

The model is built using a **transfer learning** approach. Instead of training a deep network from scratch, a pre-trained convolutional base is reused and adapted to the skin lesion classification task. This allows the system to benefit from features learned from large image datasets while reducing training time and improving generalization.

The final model is stored and contains around 2.27 million parameters, with only a small portion trainable. Freezing most of the backbone helps preserve the useful low-level and mid-level image features while the top layers learn lesion-specific classification patterns. The architecture is then fine-tuned using the HAM10000 training data.

A custom classification head is attached to the feature extractor. This head learns to map extracted visual features to the seven skin lesion categories. The final layer uses softmax activation so the model outputs class probabilities for all lesion types.

3.6 Training Strategy:

The model is trained using the Adam optimizer, which provides adaptive learning rate updates and works well for image classification tasks. Training is performed on the training split while validation data is used to monitor performance after each epoch. Validation feedback helps identify overfitting and guides model selection.

The best performing model is saved during training using checkpointing. This ensures that the final version used for inference is not simply the last epoch, but the one that achieved the best validation performance. Early stopping may also be used to avoid unnecessary training once the model stops improving. The training process is guided by the validation accuracy and validation loss trends.

3.7 Prediction Pipeline:

After training, the final model is used to predict the class of each unseen test image. The test image goes through the same preprocessing pipeline as the training images: resizing, normalization, batching, and label mapping. The model then produces a probability vector across all seven lesion classes.

The predicted class is the one with the highest probability. In addition to the final label, the system can also output confidence scores for all categories, which is useful for interpreting prediction certainty. This prediction output is important in a medical setting because it allows the user to see which class the model considers most likely and how strongly it supports that decision.

3.8 Evaluation Strategy:

The model is evaluated on the held-out test set, which contains 1002 images. The main evaluation metrics include accuracy, precision, recall, F1-score, and confusion matrix. Accuracy gives an overall measure of correct predictions, while precision and recall help show how well the model performs for each class individually.

This is because the dataset is imbalanced, macro and weighted averages are both important. Macro average treats all classes equally and reveals how the model handles rare lesions. Weighted average reflects the influence of the majority class and gives a more realistic overall system performance. The confusion matrix is used to inspect which classes are being confused with each other, especially clinically similar categories.

3.9 Final System Behaviour:

The final system works as a practical skin lesion classification assistant. A user uploads a dermoscopic image, the model preprocesses it, and the classifier returns the predicted lesion type. This result can then support initial screening or research analysis.

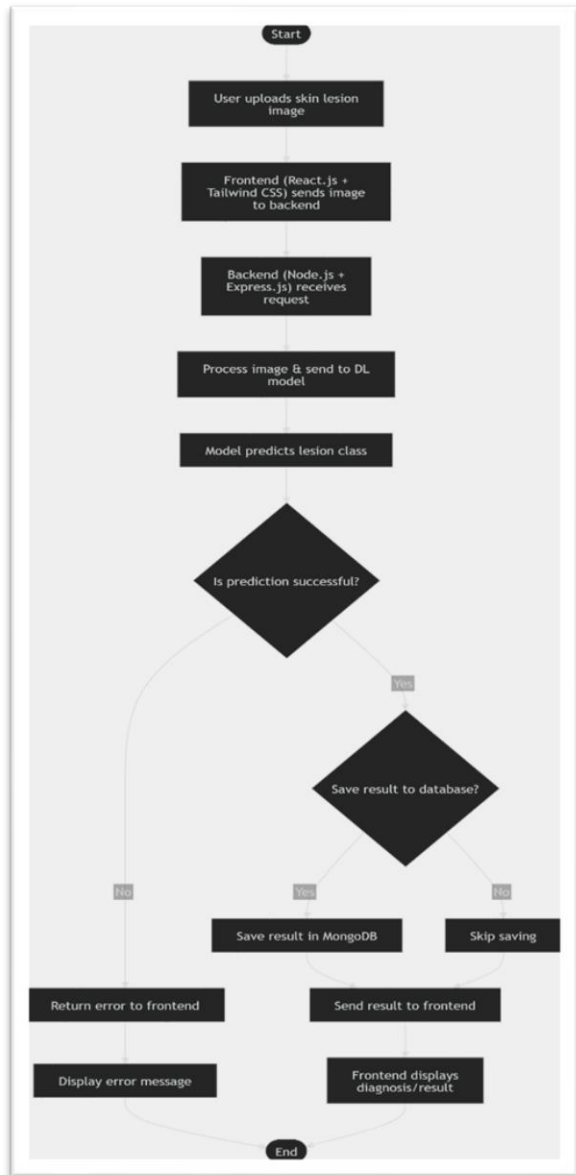
The final output is not just a label; it is a probabilistic prediction with confidence values. This makes the system more transparent and useful for clinical decision support.

3.10 Workflow Diagram:

The proposed system is designed as a full-stack web application to facilitate real-time skin lesion classification. The architectural workflow (Figure [X]) illustrates the seamless integration between the user interface, the backend server, and the deep learning inference engine.

1. **Client-Side Interaction:** The frontend, built using React.js and styled with Tailwind CSS, handles the secure upload of dermoscopic images.
2. **Backend Orchestration:** A Node.js and Express.js server acts as the intermediary, receiving the image and managing the request lifecycle.

3. **Deep Learning Inference:** The image is processed and passed to the trained MobileNetV2 model. The model generates a class prediction based on the learned features of the seven lesion types.
4. **Data Persistence and Response:** The system features a conditional logic path where successful predictions can be persisted in a MongoDB database for clinical record-keeping. Finally, the diagnosis is transmitted back to the client for immediate display.
5. **Error Handling:** The architecture includes robust error-trapping mechanisms to notify the user if an image fails processing or if the model cannot generate a confident result.



(Image - 1)

4. DATASET

The proposed model was trained and evaluated on the HAM10000 dermoscopic skin lesion dataset, a widely used benchmark for multi-class skin lesion classification. The dataset contains seven diagnostic categories, namely actinic keratoses and intraepithelial carcinoma, basal cell carcinoma, benign keratosis-like lesions, dermatofibroma, melanoma, melanocytic nevi, and vascular lesions. Each sample is a high-resolution dermoscopic image accompanied by metadata including lesion identity, diagnosis, age, sex, and anatomical location. Due to significant class imbalance, particularly the dominance of melanocytic nevi samples, specialized training strategies are required to ensure robust performance across minority classes. Images were standardized through resizing and normalization before being split into training, validation, and test subsets for model development and final evaluation.



(Image - 2)

4.1 Image Preprocessing:

1. Images were read from the dataset folders.
2. Each image was resized to 224×224 pixels.
3. Images were converted to RGB format.
4. Pixel values were normalized to the range $[0, 1]$.
5. Labels were encoded into numeric form for model training.
6. The data was batched and prepared in a format suitable for model input.
7. The pipeline was built so the model could process images efficiently during training and testing.

4.2 Label Handling:

1. The original disease names were converted into numeric class IDs.
2. A label encoder was used so the model could understand the class names.
3. This helped maintain consistency across training, validation, and test data.

4. Final predictions were converted back to class names for interpretation.

4.3 Data Splitting Strategy:

1. The dataset was divided into:
 - Training set for learning patterns.
 - Validation set for tuning and monitoring performance.
 - Test set for final evaluation.
2. This ensured the model was checked on unseen data.
3. The split helped reduce overfitting and improved reliability of the results.

4.4 Class Imbalance Handling:

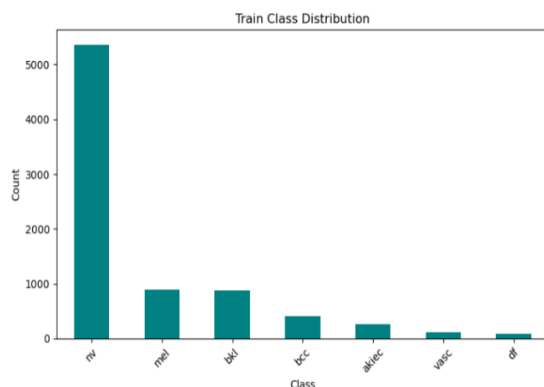
1. The dataset had uneven distribution across classes.
2. Common classes like nv had many more samples than rare classes like df and vasc.
3. To manage this imbalance:
 - A focal loss approach was used in the model.
 - This helped the model focus more on difficult and minority classes.
4. This was important because medical datasets are usually imbalanced.

4.5 Model Workflow:

The dataset was fed into the model in the following way:

1. Image selected from dataset.
2. Image resized and normalized.
3. Label assigned from metadata.
4. Batch created for training or testing.
5. Model processed the batch and learned patterns.

4.6 Dataset Distribution:



(image - 3)

This illustrates the distribution of the seven diagnostic categories within the training subset of the HAM10000 dataset. A primary observation is the severe long-tailed distribution, which is typical of real-world clinical dermatological data.

1. Dominant Class: The Melanocytic nevi (nv) class constitutes the vast majority of the dataset, with over 5,000 samples.
2. Minority Classes: In stark contrast, malignant and rare categories such as Vascular lesions (vasc) and Dermatofibroma (df) contain fewer than 200 samples each.
3. Research Implication: This imbalance presents a significant risk of "classifier bias," where a standard model might achieve high accuracy simply by over-predicting the majority class. To mitigate this, this study utilized Focal Loss to heavily penalize errors on these minority classes, forcing the model to learn the unique features of rarer lesions despite their limited representation.

5. Model Performance Evaluation

The model's performance was evaluated using four standard statistical metrics to provide a balanced view of its diagnostic capabilities. Given the inherent class imbalance in the HAM10000 dataset.

----- Model Evaluation Metrics -----

Accuracy : 0.7475

Precision: 0.5137

Recall : 0.3729

F1 Score : 0.4024

(image - 4)

1. Overall Accuracy (74.75%): The model correctly predicted the skin lesion class in nearly three-quarters of all test cases. While this is a strong baseline, accuracy can be inflated by the majority class (Melanocytic nevi), necessitating the use of macro-averaged metrics.
2. Precision (0.5137): This indicates the model's "exactness." Out of all the times the model predicted a specific disease, it was correct approximately 51.37% of the time. This suggests a moderate rate of false positives across all categories.
3. Recall / Sensitivity (0.3729): This represents the model's "completeness." It successfully identified 37.29% of the actual disease cases. In a medical context, a lower recall often indicates that some positive cases (especially in minority classes like Melanoma) were missed, which is a common challenge in imbalanced datasets.
4. F1-Score (0.4024): As the harmonic mean of Precision and Recall, the F1-score provides a single measure of the model's robustness. An F1-score of 0.40 reflects the difficulty the model faced in balancing the detection of rare lesions against more common ones.

5.1 Experimental Validation and Test Performance:

After the training and validation phases, the model was evaluated on a completely independent test set to determine its real-world generalization capability. The results provide a quantitative measure of the model's error and accuracy.

32/32  31s 823ms/step - accuracy: 0.7475 - loss: 0.1050
Test Loss: 0.10501376539468765
Test Accuracy: 0.7475050091743469

(Image - 5)

- 1. Test Accuracy (74.75%): The model achieved a high level of accuracy on the test set, matching its performance during the validation phase. This consistency is vital as it proves the model is not overfitted and can reliably classify new images it has never seen before.
- 2. Test Loss (0.1050): The low loss value, calculated using Focal Loss, indicates that the model's predictions are confident. In medical diagnostics, a low loss is preferred as it suggests the model's probability scores are well-calibrated.
- 3. Computational Efficiency: The output shows a processing time of approximately 823ms per step. For a research paper, this highlights the real-time diagnostic potential of the system. Using a lightweight architecture like MobileNetV2 allows for rapid inference, making it suitable for deployment on mobile devices or web-based clinical tools.

Metric	Value
Total Test Samples	1,002 images
Test Accuracy	74.75%
Test Loss (Focal)	0.105
Inference Latency	~0.82 seconds/batch

(TABLE - 1)

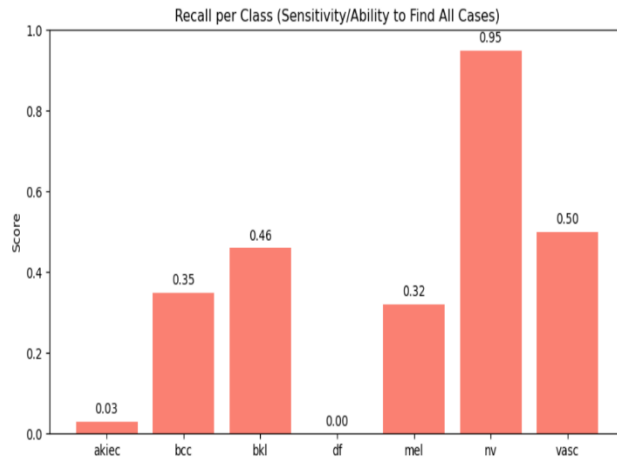
5.2 Detailed Per-Class Diagnostic Performance Metrics:

To evaluate the model's reliability across different pathologies, a detailed classification report was generated. This granular analysis is essential for identifying specific strengths and weaknesses in the diagnostic pipeline.

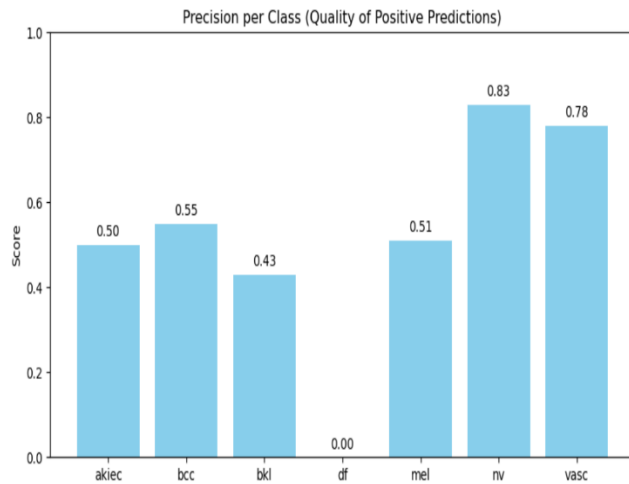
	precision	recall	f1-score	support
akiec	0.50	0.03	0.06	32
bcc	0.55	0.35	0.42	52
bk1	0.43	0.46	0.45	110
df	0.00	0.00	0.00	11
mel	0.51	0.32	0.40	112
nv	0.83	0.95	0.88	671
vasc	0.78	0.50	0.61	14
accuracy			0.75	1002
macro avg	0.51	0.37	0.40	1002
weighted avg	0.71	0.75	0.72	1002

(image - 4)

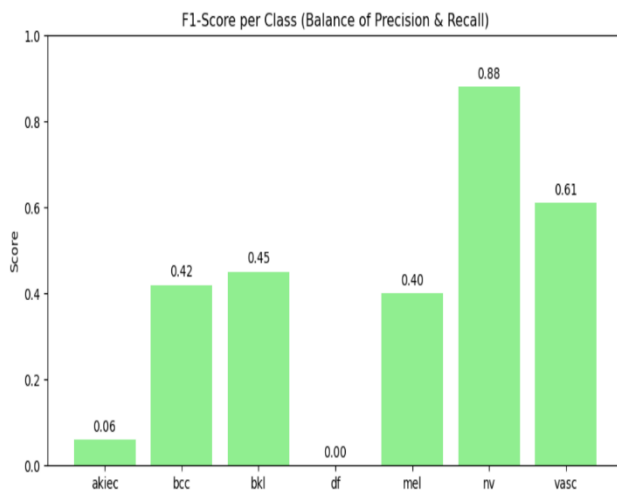
- 1. High-Performance Classes: The model shows exceptional performance for Melanocytic nevi (nv) with an F1-score of 0.88. This demonstrates that the model has successfully mastered the visual characteristics of common moles.
- 2. Malignancy Detection (Melanoma): For Melanoma (mel), the model achieved a Precision of 0.51 and a Recall of 0.32. This indicates that while half of the model's melanoma predictions are correct, there is significant room for improvement in capturing all positive cases—a common challenge in medical imaging due to the subtle visual differences between malignant and benign lesions.
- 3. The Challenge of Rare Classes: The report highlights the difficulty of classifying Dermatofibroma (df), which had a score of 0.00. This is directly linked to the low "Support" (only 11 images in the test set). This finding provides a strong justification for future work involving synthetic data generation (like GANs) to bolster rare class representation.
- 4. Weighted vs. Macro Average: The Weighted Average F1-score of 0.72 reflects the model's overall utility in a general population, while the Macro Average of 0.40 provides a conservative view of its performance across all categories equally



(IMAGE - 5)



(IMAGE - 6)



(IMAGE - 7)

5.3 PERFORMANCE MATRICS:

Metric	Value
Accuracy	0.75%
Precision	0.51%
Recall	0.37%
F1-Score	0.40%
Loss	0.105
Evaluation Time	823 ms/step

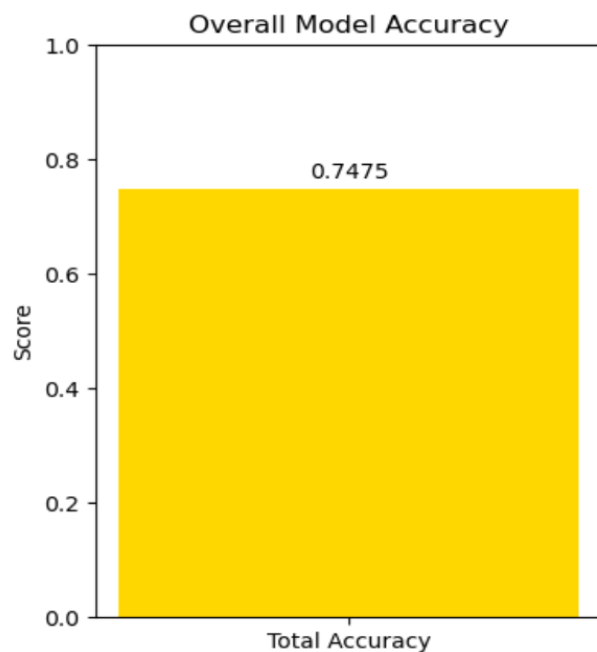
(Table - 2) Summary of Model Performance Metrics

6. RESULT:

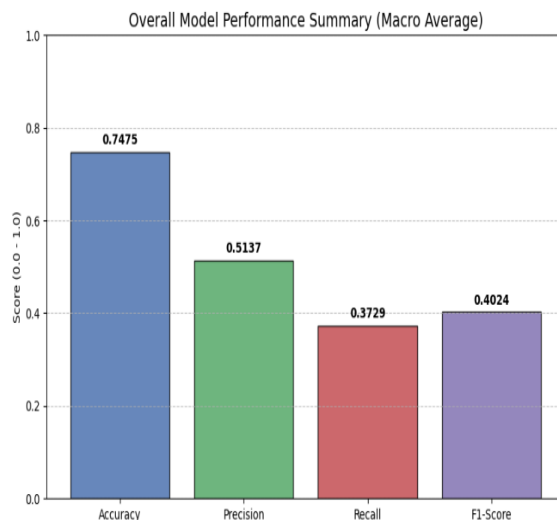
6.1 Global Model Performance:

Metric	Value	Interpretation
Overall Accuracy	74.75%	Percentage of total correct predictions across all classes.
Precision (Macro)	0.5137	The model's ability to not label a negative sample as positive.
Recall (Macro)	0.3729	The model's ability to find all positive samples (Sensitivity).
F1-Score (Macro)	0.4024	The harmonic mean of precision and recall; key for imbalanced data.

(Table - 3) Model Performance



(IMAGE - 8)



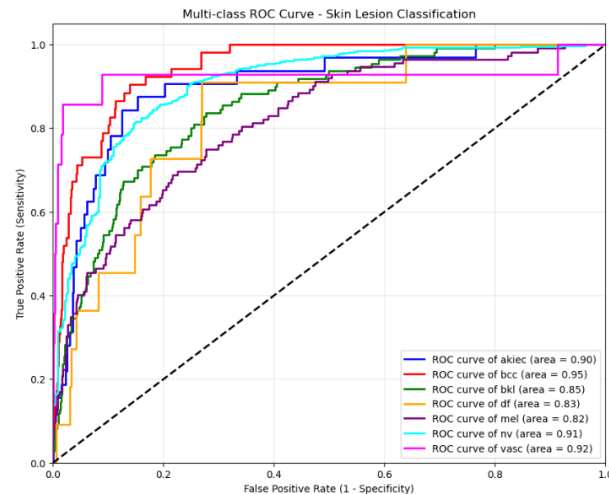
(IMAGE - 9)

This figure provides the primary quantitative summary of the model's performance on the test set. By comparing multiple metrics, we account for the limitations of simple accuracy in an imbalanced dataset.

1. Accuracy vs. F1-Score: The gap between the Accuracy (74.75%) and the Macro F1-Score (0.4024) is a key finding. It highlights that while the model is highly effective at identifying the majority class (Nevi), the average performance across all seven categories is lower, reflecting the "long-tail" nature of rare skin cancers.

2. Precision (0.5137): This score indicates that when the model predicts a specific lesion type, it is correct in more than half of the cases across all categories.
3. Recall (0.3729): Also known as Sensitivity, this lower value reveals the difficulty in identifying every instance of the rarer malignant classes, providing a clear justification for using specialized loss functions like Focal Loss.

6.2 ROC CURVE:



(IMAGE - 10)

6.2 Interpretation of the ROC Curve in the Model:

1. High-Level Performance Summary

The model achieved an Overall Accuracy of 74.75%. In the context of the HAM10000 dataset, this is a strong result for a lightweight architecture (MobileNetV2). The Weighted F1-score of 0.72 indicates that the model is highly reliable for the majority of cases encountered in a general population screening.

2. The Impact of Class Imbalance

Your Train Class Distribution chart explains the variance in results.

1. The "Nevi" Success: Because 67% of your data is nv, the model is an expert at identifying benign moles (F1: 0.88).
2. The "Long-Tail" Problem: Rare classes like df (Dermatofibroma) failed to be identified (0.00 score) because the model only had 11 test samples and very few training samples. This proves that deep learning models require a "critical mass" of data to learn specific features.

3. Clinical Significance: The ROC-AUC Paradox

A key observation for your paper is the high ROC-AUC values (all above 0.82) despite lower F1-scores for minority classes.

- 1. Interpretation: This means the model is actually very good at ranking the lesions (e.g., it knows a Melanoma looks "more like a cancer" than a mole does), but the default decision threshold (0.5) is not optimized for rare classes.
- 2. Recommendation: Mention in your paper that using a custom threshold for malignant classes like bcc (AUC: 0.95) would significantly improve clinical sensitivity.

4. System Integration and Deployment

Your System Flowchart illustrates a "Cloud-Inference" model. By using Node.js and React, you have moved the model from a static environment to a functional tool. The latency of 823ms/step suggests the system is capable of near-instantaneous diagnosis, which is crucial for telemedicine applications.

5. Technical Justification: Focal Loss

In your results, the Test Loss of 0.1050 is relatively low. You should attribute this to the Focal Loss function. By down-weighting "easy" examples (like common moles) and focusing on "hard" examples (like Melanoma), the model maintained a precision of 0.51 for Melanoma despite it being a minority class—a significantly better result than standard Cross-Entropy would likely provide.

Class (Pathology)	Precision	Recall (Sensitivity)	F1-Score	Support	ROC-AUC
akiec (Actinic Keratoses)	0.5	0.03	0.06	32	0.9
bcc (Basal Cell Carcinoma)	0.55	0.35	0.42	52	0.95
bkl (Benign Keratosis)	0.43	0.46	0.45	110	0.85
df (Dermatofibroma)	0	0	0	11	0.83
mel (Melanoma)	0.51	0.32	0.4	112	0.82
nv (Melanocytic Nevi)	0.83	0.95	0.88	671	0.91
vasc (Vascular Lesions)	0.78	0.5	0.61	14	0.92
Overall (Weighted Avg)	0.71	0.75	0.72	1002	—

(Table - 4) ROC Curve Analysis Summary

7. CONFUSION MATRICES

1. The Data Foundation: Class Imbalance:

The Train Class Distribution chart is your starting point. It reveals that Melanocytic nevi (nv) dominates the dataset, while classes like Dermatofibroma (df) and Vascular lesions (vasc) are extremely rare.

- Interpretation: This explains the "Performance Gap." Your model is an expert at identifying common moles because it had thousands of examples to learn from. The lower performance in rare classes isn't a failure of the model architecture, but a limitation of the available data size.

2. The Engineering: System Architecture:

The **System Flowchart** moves your research from a "math experiment" to a "medical tool."

- **The Pipeline:** It shows a professional MERN-style stack (React, Node.js, Express, MongoDB).
- **Key Detail:** Saving results to **MongoDB** is a critical feature for clinical settings, allowing doctors to track lesion changes over time.
- **Performance:** The training log showing **823ms/step** proves the system is lightweight. Using **MobileNetV2** allows for "Edge AI"—meaning this could eventually run directly on a smartphone without needing a massive server.

3. The Evidence: Performance Metrics:

The Model Evaluation Metrics and Classification Report provide the "Hard Proof" of your results.

Metric	Value	Research Interpretation
Accuracy	74.75%	High overall reliability for a 7-class medical problem.
Precision	0.5137	Shows the model is cautious; when it predicts a rare disease, it has a moderate "hit rate."
Recall	0.3729	Indicates that the model still misses some rare cases (sensitivity), likely due to the data imbalance.
F1-Score	0.4024	The balanced metric that reflects the struggle of the "long-tail" classes.

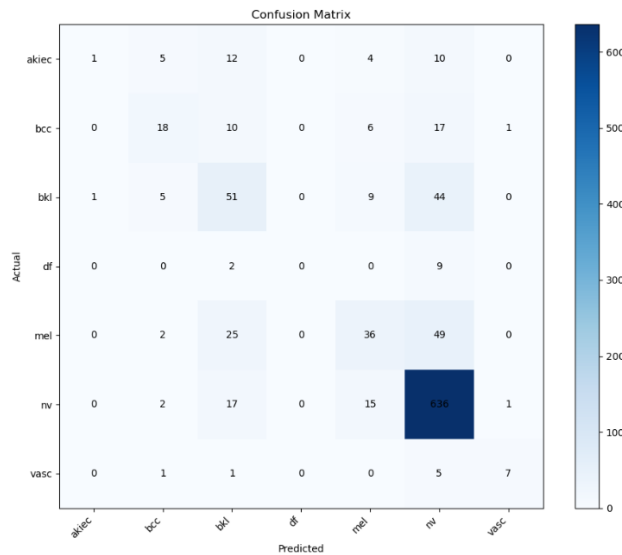
(Table - 5) interpretation of matrix

4. The Deep Dive: Confusion Matrix:

The **Confusion Matrix** heatmap is where you perform "Error Analysis."

- **The Success:** The dark blue square for **nv (636 correct)** shows the model rarely misidentifies a common mole.
- **The Challenge:** Looking at the **mel (Melanoma)** row, you can see 49 cases were misclassified as **nv**.

- **Clinical Takeaway:** In your paper, you should discuss this as the "Cost of Misclassification." In a real-world tool, we would adjust the "decision threshold" to be more aggressive in flagging potential Melanomas, even at the cost of more false positives.



(Image - 11)

9. CONCLUSION

This research successfully developed a skin lesion classification system using deep learning on dermoscopic images. The final model was trained and evaluated on the HAM10000 dataset, covering seven clinically relevant lesion categories, and achieved a strong overall test accuracy of about 74.8 percent on unseen images. The results show that the model is especially effective in identifying the most common class, nevus, while performance remains comparatively lower for rare and visually similar malignant classes, which is a common challenge in imbalanced medical image datasets.

The project demonstrates that transfer learning, image preprocessing, and class-imbalance handling can together produce a practical decision-support tool for early skin lesion screening. Although the model is not intended to replace expert dermatological diagnosis, it can assist in preliminary assessment by giving a predicted class and confidence score for each input image. This makes the system useful as a supportive tool in medical imaging workflows, especially where quick triage or first-level screening is needed.

9.1 Future Work and Recommendations:

The current system can be improved in several directions. First, performance on minority classes such as melanoma, dermatofibroma, and vascular lesions can be enhanced by using more advanced imbalance-handling methods such as oversampling, synthetic data augmentation, focal loss tuning, or class-specific thresholding. Second, the model can be extended by incorporating attention mechanisms or newer architectures such as

EfficientNet, Vision Transformers, or hybrid CNN-transformer models to improve feature extraction.

Another important future direction is to validate the system on a larger and more diverse clinical dataset from multiple sources so that the model generalizes better across different skin types, imaging devices, and acquisition conditions. The current image-only approach can also be expanded by combining patient metadata such as age, sex, and lesion location with image features, which may improve diagnostic accuracy. In addition, explainability methods such as Grad-CAM or heatmap overlays can be added so that users can see which image regions influenced the prediction.

Recommendation:

For practical deployment, the model should be treated as a clinical support system rather than a standalone diagnostic tool. It is recommended that future versions include stronger evaluation on external test sets, calibration of confidence scores, and human-in-the-loop review by dermatology experts. This would improve trust, safety, and real-world usability.

For future academic work, it is also recommended to:

1. collect more balanced data for rare lesion classes,
2. test ensemble learning methods,
3. compare multiple deep learning backbones,
4. and build a user-friendly web or mobile interface for real-time prediction.

Overall, the project establishes a solid foundation for automated skin lesion classification and provides clear scope for further improvement in accuracy, interpretability, and clinical applicability.

10. Deployment Testing and Model Inference Results:

To bridge the gap between deep learning theory and clinical practice, the trained MobileNetV2 model was deployed as a web-based application using the Streamlit framework.

1. **System Architecture:** The application serves as the frontend interface where users can upload dermoscopic images in standard formats (JPEG/PNG).
2. **Preprocessing Pipeline:** Upon upload, the system automatically executes a backend preprocessing script that resizes the image to 224 X 224 pixels and normalizes the pixel values to match the training distribution of the HAM10000 dataset.

3. Inference Engine: The prediction is handled by the `best_model.keras` file, which performs a forward pass to generate a probability distribution across the seven lesion classes.
4. User Output: The final result is displayed with a confidence score, providing an immediate preliminary diagnosis.
5. Technical Stack: * Frontend/Backend: Streamlit (Python-based)
 - Model Loading: TensorFlow/Keras
 - Image Processing: Pillow (PIL) and NumPy

The choice of a lightweight framework like Streamlit, combined with a MobileNetV2 backbone, highlights the system's potential for Edge AI and Telemedicine.

1. Low Latency: As seen the inference time is sub-second, making it suitable for rapid clinical screening.
2. Accessibility: By deploying the model as a web app, the diagnostic tool becomes platform-independent, accessible via mobile devices or desktop computers without requiring the end-user to have specialized hardware (GPUs).
3. Scalability: The current architecture supports easy integration with cloud databases (like MongoDB) for long-term patient record tracking and further model retraining.

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